# An Interleaved Forward/Flyback DC–DC Converter With Bidirectional Capability and Wide Voltage Range Characteristic

Robson Mayer, Luan Souza de Oliveira, Menaouar Berrehil El Kattel, Luiz Carlos Gili, and Sérgio Vidal Garcia Oliveira, *Senior Member, IEEE*

***Abstract*—This article introduces an interleaved forward/flyback converter designed to achieve high power capability, bidirectional operation, galvanic isolation, and low ripple current. By integrating a coupled inductor and utilizing interleaved operation, the circuit achieves a broad range of voltage gains while effectively reducing ripple current at the switching frequency by a factor of two. The design employs an interleaved two-switch forward topology that operates bidirectionally while requiring fewer switches than conventional dual-active-bridge converters, making it particularly well-suited for applications such as rechargeable energy storage systems. The interleaved structure not only minimizes the size of passive filters but also accommodates a wide range of voltage variations. In addition, the converter operates with zero-voltage switching in forward mode, significantly reducing losses and electromagnetic interference, thereby enhancing overall efficiency. This article provides a detailed examination of the converter's operational principles, including the behavior of inductor currents in continuous, discontinuous, and boundary conduction modes. Furthermore, a comprehensive circuit analysis is presented, outlining voltage and current stresses under both power flow directions. A comparative performance evaluation with similar isolated converters highlights the key advantages and potential limitations of the proposed topology. Finally, experimental validation is conducted using a 3.5 kW prototype, demonstrating an efficiency exceeding 90% in both power flow directions.**

Received 25 November 2024; revised 25 February 2025; accepted 13 March 2025. Date of publication 20 March 2025; date of current version 26 May 2025. This work was supported in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior–CAPES–Brazil (PROAP/AUXPE DS) under Grant 1928/2023 and Grant 88881.898694/2023-01, in part by the National Council for Scientific and Technological Development (CNPq) under Grant 303460/2023-7, and in part by the Santa Catarina State Research and Innovation Support Foundation (FAPESC), under Process 2023TR00924, all in Brazil. Recommended for publication by Associate Editor E. S. Lee. *(Corresponding author: Robson Mayer.)*

Robson Mayer is with the Faculdade de Engenharia Elétrica e de Computação da UNICAMP, Universidade Estadual de Campinas, Campinas 13083-852, SP, Brazil (e-mail: rmayer@unicamp.br).

Luan Souza de Oliveira and Luiz Carlos Gili are with the Núcleo de Processamento de Energia Elétrica, Universidade do Estado de Santa Catarina (UDESC), Joinville 89219-710, SC, Brazil (e-mail: luan.oliveira96@edu.udesc.br; Luiz.c.gili@nidec-ga.com).

Menaouar Berrehil El Kattel is with the Universidade Federal do Ceará (UFC), Campus do Pici. Av. Humberto Monte, Joinville 60455-760, SC, Brazil (e-mail: berrehilelkattel@dee.ufc.br).

Sérgio Vidal Garcia Oliveira is with the Centro de Ciências Tecnológicas, Departamento de Engenharia Elétrica, Universidade do Estado de Santa Catarina (UDESC), Joinville 89223-100, SC, Brazil (e-mail: sergio_vidal@ieee.org).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TPEL.2025.3553280.

Digital Object Identifier 10.1109/TPEL.2025.3553280

***Index Terms*—Bidiretional isolated converter, electric vehicle (EV) charging, forward/flyback topology, interleaved operation.**

## I. Introduction

DURING the integration of renewable energy into electrical systems, distributed generation (DG) has emerged, focusing on systems for mini- and microgeneration [1]. With DG becoming more common in power systems, novel methods of processing power are essential. Electric vehicles (EVs) play an important role in power systems, acting as both consumers and suppliers [2].

The integration of EVs into smart grids represents a rapidly expanding and highly promising area of research. To fully realize the potential benefits, it is crucial to carefully examine the economic, structural, technological, and integration dimensions [3]. This analysis explores a variety of potential approaches to overcome implementation challenges and identifies key areas for further investigation to develop effective solutions to these complex issues.

EVs can improve smart grids by storing excess renewable energy and supplying it during peak demand [4]. Their bidirectional charging capabilities allow EVs to use batteries to improve the flexibility and efficiency of the grid. of the grid. This integration supports a more resilient and sustainable energy infrastructure [5].

Recent research has focused on the development of dc–dc converters in integration of rechargeable energy storage system (RESS) aiming to enhance energy management, storage capabilities, and their application in dc microgrids [6]. Traditional dc–dc converters, such as buck and boost converters, have been widely used in various integrations due to their simplicity and effectiveness. Although significant progress has been made in the field of dc–dc converters, there remain notable challenges in designing converters capable of seamlessly integrating diverse power sources, offering galvanic isolation, and sustaining high efficiency across a wide range of load and voltage conditions.

Dual-active-bridge (DAB) converters are highly suitable for EVs RESS due to their bidirectional power flow capability, high efficiency through zero-voltage switching (ZVS), galvanic isolation, and simple structure with minimal components. They effectively manage energy transfer between the battery and the

0885-8993 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

grid, ensuring efficient charging and discharging. Furthermore, DAB converters offer dynamic control over power flow, improving overall system performance and reliability [7]. In order to achieve the high voltage gain and reduce the switch voltage stress of the converters, coupled inductors can be used. However, DAB converters also present certain limitations and challenges that require careful consideration. These include the need for advanced control strategies to effectively manage power flow, particularly under light-load conditions or wide input/output voltage variations (e.g., triple phase shift control) [8]. In addition, issues such as high circulating currents, a limited ZVS range—especially under light loads or significant input/output voltage mismatches—and sensitivity to variations in transformer leakage inductance must be addressed [9]. Furthermore, achieving an optimal balance between efficiency, power density, and cost remains a significant challenge in the design of DAB converters [10].

While DAB converters present notable advantages for EV and RESS applications, addressing their limitations necessitates thoughtful design considerations, such as the development of advanced control strategies, careful component selection, and the implementation of robust thermal management techniques. Ongoing research efforts are actively exploring innovative solutions to mitigate these challenges. For example, a novel topology derived from the two-transformer forward interleaved configuration [11], [12] has been proposed to meet the stringent requirements for bidirectional converters, as previously highlighted for DAB. Although the topology proposed in [11] utilizes two transformers connected in series, variations in their construction result in an imbalance in magnetic flux across the magnetizing inductance. This leads to a limited voltage gain and a constrained ZVS range. In addition, the topology lacks the capability for bidirectional power flow, which further restricts its versatility in practical applications.

Classical hard-switched flyback converters can achieve high voltage gain by employing a high turns ratio in the coupled inductors [13]. However, with the increase of the turns ratio, the volume of the magnetic core becomes much larger, the copper windings need to be longer, and the stress on the switches is high. In addition, the voltage stress of the output diode will also become very large, and the conduction loss of the switch and the copper loss of the windings increase, which induces low efficiency performance [14]. The flyback converter is recognized as one of the most cost-effective isolated topologies due to its minimal component count, which allows the integration of the energy storage inductor with the transformer. However, the overall cost is influenced not only by the topology itself but also by the specific implementation. As a result, not all implementations of the flyback topology necessarily result in a low-cost converter. Research in the literature has explored the use of conventional interleaved flyback converters and multiwinding configurations to enhance power distribution, efficiency, and voltage gain in isolated applications, particularly for renewable energy and high-power distribution systems [15], [16]. However, these configurations are primarily limited to unidirectional power transfer, which restricts their applicability in scenarios requiring bidirectional energy flow

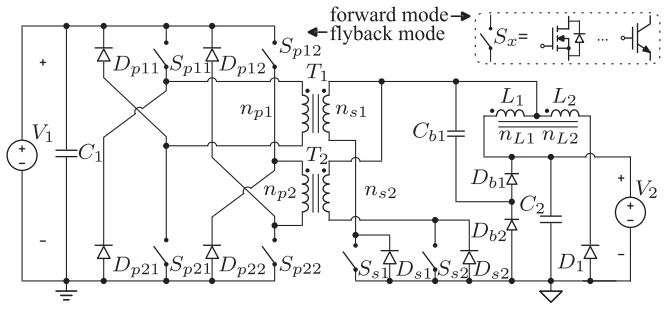



Fig. 1. Proposed bidirectional interleaved forward/flyback converter.

This study introduces and elaborates on the development of a bidirectional interleaved forward/flyback converter, derived from the two-switch forward converter topology, designed for high-power applications. The proposed system incorporates an interleaved flyback topology on the secondary side of the transformers, as illustrated in the circuit diagram provided in Fig. 1. In both operating directions, the circuits are designed to operate 180° out of phase with each other. In forward mode, the peak voltage across each switch is clamped to the dc voltage source, and the current stresses are comparable to those of a DAB converter. However, a limitation in forward mode is that the transformer flux swing operates unidirectionally, necessitating a larger transformer core compared to the full-bridge configuration. In reverse mode, the maximum power transfer is constrained by the selected magnetizing inductance value. That said, a reduced magnetizing inductance extends the voltage gain range in flyback mode, effectively mitigating this limitation. In addition, the topology eliminates the need for flux balancing methods, such as a blocking capacitor, further simplifying the design.

In conclusion, this topology integrates a coupled inductor $L_1$, $L_2$ ($L_a$) and interleaved transformers $T_1$, $T_2$, preserving similar operational characteristics while utilizing fewer semiconductor devices. By doubling the switching frequency in the coupled inductor, the design reduces the overall volume of inductors. In forward mode, the transformers provide galvanic isolation, voltage gain, and current source characteristics at the output. In reverse mode, the converter takes advantage of an interleaved flyback topology, including high voltage gain, reduced magnetic volume, a simplified circuit structure, and current-fed capabilities. The proposed converter maintains a continuous operation with a constant and low-ripple current at $V_2$, enabled by the coupled inductor $L_1$. This feature makes it particularly well-suited to meet the demands of modern battery systems. We also regard this outcome as a significant research contribution, as this technology has the potential to facilitate the development of various applications where cost-effectiveness and simplicity are key considerations.

## II. Proposed Bidirectional DC–DC Converter Interleaved Forward/Flyback

This article studies a bidirectional interleaved forward/flyback converter, as drawn in Fig. 1. The behavior analysis is conducted

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

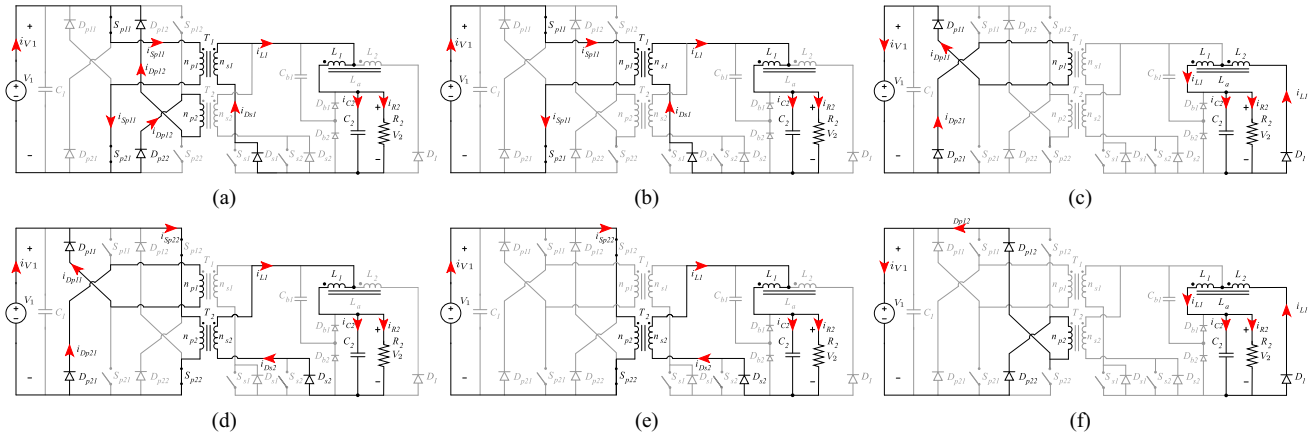



Fig. 2. Steps of forward in CCM. (a) Stage I $[t_0 - t_1']$, (b) stage II $[t_1' - t_1]$, (c) stage III $[t_1 - t_2]$, (d) stage IV $[t_2 - t_2']$, (e) stage V $[t_2' - t_3]$, and (f) stage VI $[t_3 - t_4]$.

in three modes, continuous conduction mode (CCM), discontinuous conduction mode (DCM), and boundary conduction mode (BCM). The proposed converter is an isolated bidirectional forward/flyback converter. In forward mode, energy flows from $V_1$ to $V_2$, while in flyback mode, energy flows from $V_2$ to $V_1$.

In forward mode, the converter operates as a step-down converter with a duty cycle range of $0 < D < 0.5$, which is limited to half the switching period and typically used in two-phase systems. The transformers operate 180° out of phase, and the output behaves as a current source, allowing its ripple to be adjusted based on the specific requirements of the application. The peak voltage stress on each switch remains limited to the dc voltage source $V_1$, while the current stresses are comparable to those observed in the DAB converter. Furthermore, no additional flux balancing methods, such as a blocking capacitor, are required.

In flyback mode, the same components are employed, however, the topology operates as a step-up converter, similar to the interleaved flyback converter. In this configuration, the magnetizing inductance of the transformers facilitates energy transfer, allowing them to function as coupled inductors while ensuring galvanic isolation. In addition, a low-pass filter, consisting of components $L_1$ and $C_{b1}$, is integrated between the interleaved transformers and the source $V_2$. This filter imparts characteristics of the current source, which makes it especially suitable for applications involving emerging battery technologies.

## III. Operations Analysis of Forward in CCM

Two phases are chosen for the converter's presentation, analysis, and validation. Each phase has two switches and one transformer, so the number of transformers equals the number of branches, while the number of switches is doubled. Pulsewidth modulation (PWM) is applied in the converter, with switches on the same primary winding controlled simultaneously, but shifted by 180° relative to switches on the other windings.

The current through the coupled inductor $L_1$ never reaches zero, as the source $V_1$ supplies energy to the system intermittently—only during the conduction state of the switches.

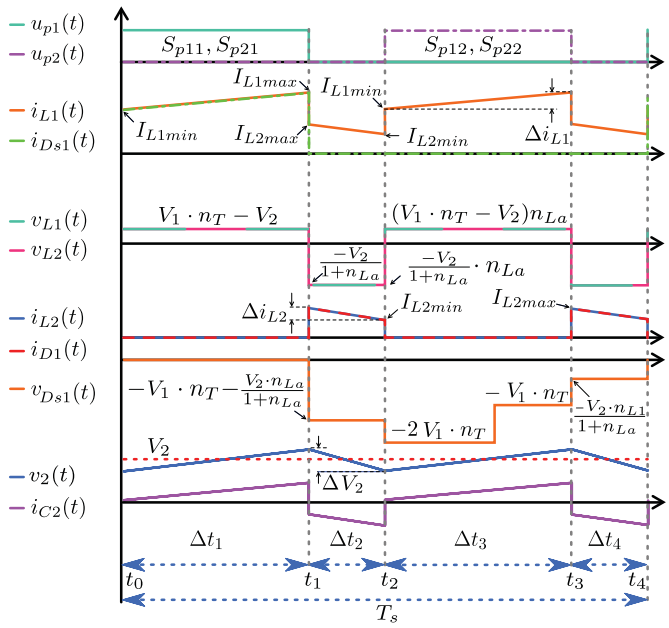



Fig. 3. Idealized waveforms of forward mode in CCM.

In this configuration, $V_2$ represents the circuit's load, with $R_2$ used in its place. In practical applications, this load corresponds to the RESS, whereas in simulations and prototype testing, a resistive load is employed to validate and analyze circuit performance under nominal conditions.

Fig. 2 illustrates six distinct substeps. In the first and fourth steps, the magnetizing inductance demagnetizes on the primary side. While these steps are essential for comprehending the converter's operation, they can be disregarded when focusing on the energy transfer balance between the input and output.

Fig. 3 illustrates the waveform of the converter operating in CCM. The static gain for $V_2$ to $V_1$ can be derived using mean voltage across $L_1$, as shown in (1). The mean voltage across $L_1$ can be expressed as an integral of this voltage over the period $T_s$ divided by 2, since the voltage across this inductor is twice

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

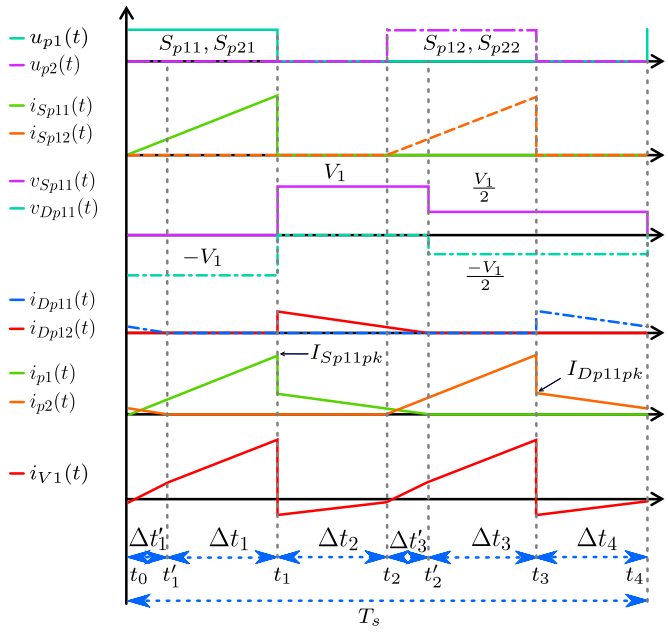



Fig. 4. Idealized waveforms of forward mode in DCM.

switching frequency

$$\overline{V_{L1}}(t) = \frac{2}{T_s}\left[\int_0^{\Delta t_1}(V_1 \cdot n_T - V_2)\,dt + \int_0^{\Delta t_2}\left(-\frac{V_2}{1+n_{La}}\right)dt\right]. \tag{1}$$

Assuming that the mean voltage across $L_1$ is zero, the mean voltage in $L_1$ can be manipulated to obtain $Q_{1\text{CCM}}$, which represents the static gain for CCM, as

$$Q_{1\text{CCM}} = \frac{V_2}{V_1} = \frac{2\,D \cdot n_T(n_{La}+1)}{2\,D \cdot n_{La}+1}. \tag{2}$$

The ratios between the windings are defined as

$$n_T = \frac{n_{s1}}{n_{p1}} = \frac{n_{s2}}{n_{p2}}, \quad n_{La} = \frac{n_{L2}}{n_{L1}} = \sqrt{\frac{L_2}{L_1}}. \tag{3}$$

By examining the waveform of the current in the output filter inductor, as depicted in Fig. 3, the required value for $L_1$ can be determined based on the desired ripple and the specified $n_{La}$ ratio, as outlined in the following equation:

$$L_1 = \frac{V_2}{\Delta I_{L1} \cdot f_s} \cdot \frac{(1-2\,D)}{2\,(n_{La}+1)}. \tag{4}$$

Fig. 5(a) illustrates the static gain in forward mode for CCM condition as a function of $D$, compared with forward classic, interleaved, and variations of $n_{La}$. In Fig. 5(b), the static gain in forward mode for DCM is presented as a function of $D$, and for multiples values of parameterized current on the load $\overline{i_{R2}}$. The corresponding voltage gain curves are plotted in Fig. 5(c), highlighting the static gain as a function of the parameterized load current $\overline{i_{R2}}$, and multiple values of $D$. This plot illustrates the operating regions and their respective boundaries.

## IV. Operations Analysis of Forward in DCM

In DCM, the current through the inductor $L_1$ drops to zero before completing half of a switching period, interrupting energy transfer from the source to the load, the idealized waveforms are presented in Fig. 4. During this time, the capacitor connected in parallel with the load supplies the necessary power. This mode is divided into six operational stages on the secondary side within one switching period. Some stages share topological states with CCM, but the key difference lies in the absence of inductor current during specific intervals. This cyclical process ensures that the load is consistently powered, even when the inductor current is zero. As seen in Fig. 6.

Fig. 5(b) illustrates the static gain as a function of $D$ and shows multiple curves for parameterized load current $\overline{i_{R2}}$.

For DCM converters, the static gain is determined based on the energy supplied by the source and the energy consumed by the load. The current waveforms shown in Fig. 4 are used to analyze the system's energy balance. The current ripple is influenced by the demagnetization time of $L_1$ and is given by

$$\Delta I_{L1} = \frac{V_1 \cdot n_T - V_2}{L_1} \cdot t_{d_{L1}}. \tag{5}$$

Demagnetization time is an important variable for DCM, and is derived as

$$t_{d_{L1}} = \frac{L_1 \cdot \Delta I_{L1} \cdot (n_{La}+1)}{V_2}. \tag{6}$$

The current ripple in $L_2$ depends on transformation ratio $n_{La}$, and is given as

$$\Delta I_{L2} = \frac{\Delta I_{L1}}{n_{La}+1}. \tag{7}$$

The energy delivered to the load is

$$W_{V2} = \frac{V_1 \cdot V_2 \cdot D^2 \cdot T_s^2}{L_1} \cdot \left(n_T - \frac{E_2}{V_1}\right) \cdot \left[1 + \left(\frac{E_1}{V_2} \cdot n_T - 1\right)\right]. \tag{8}$$

Respecting the energy conservation law, the static gain $Q_{1\text{DCM}}$ is calculated based on the parameterized current in the load, given as

$$Q_{1\text{DCM}} = \frac{D^2 \cdot n_T^2}{n_T \cdot D^2 + \overline{i_{R2}}}. \tag{9}$$

The parameterized current in the load $I_{R2}$ is expressed as

$$\overline{i_{R2}} = \frac{I_{R2} \cdot L_1 \cdot f_s}{V_1}. \tag{10}$$

Fig. 5(b) presents the static gain as a function of $D$, including different plots for the parameterized current $\overline{i_{R2}}$. As expected in DCM, the gain varies with changes in the current supplied to the load. To maintain operation in DCM, the current must be limited. When the current reaches its maximum value, the converter transitions to either BCM or CCM. This maximum current value is given by

$$I_{L1\text{BCM}} = I_{R2_{\text{avg}}} = \frac{\Delta I_{L1}}{2} = \frac{V_2}{L_1 \cdot f_s} \cdot \frac{(1-2\,D)}{4\,(n_{La}+1)}. \tag{11}$$

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

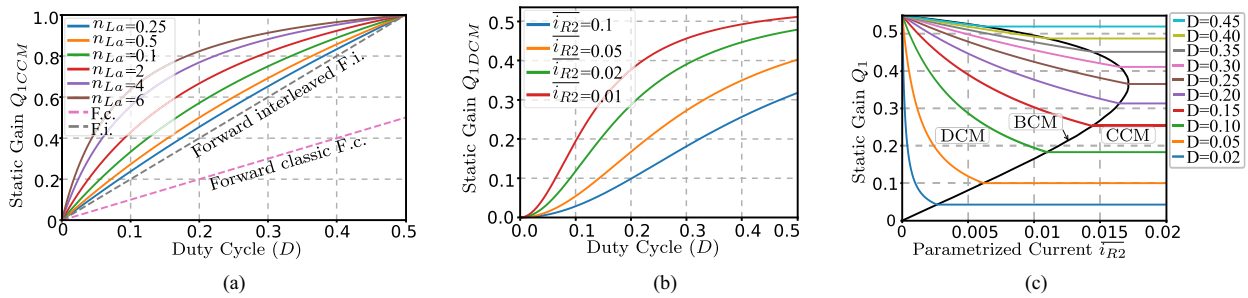



Fig. 5. Static gain of forward mode. (a) CCM. (b) DCM. (c) Output characteristics.

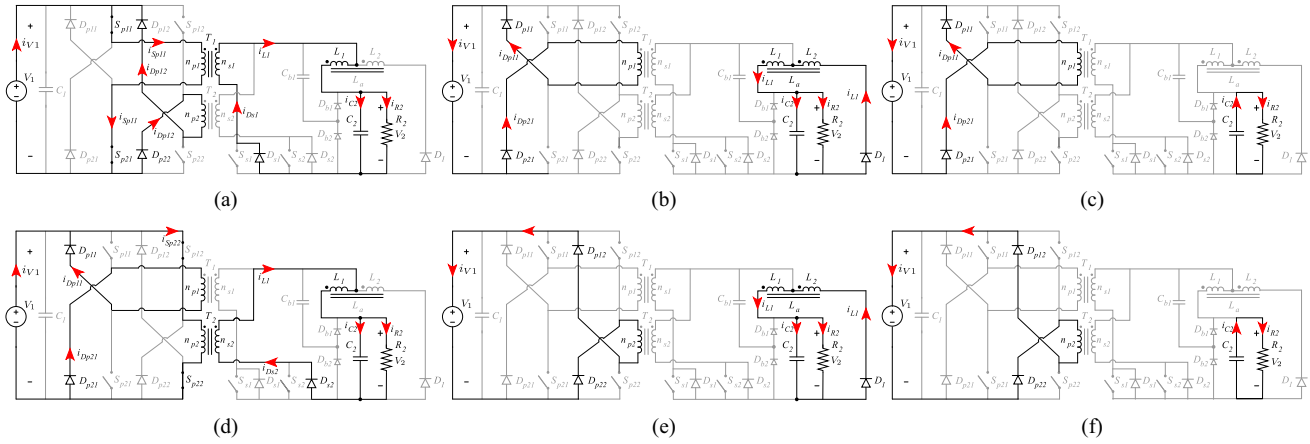



Fig. 6. Steps of forward in DCM. (a) stage I $[t_0 - t_1']$, (b) stage II $[t_1' - t_1]$, (c) stage III $[t_1 - t_2]$, (d) stage IV $[t_2 - t_2']$, (e) stage V $[t_2' - t_3]$, and (f) stage VI $[t_3 - t_4]$.

And maximum load is expressed as

$$R_{2\text{BCM}} = \frac{4\,L_1 \cdot f_s \cdot (n_{La}+1)}{(1-2\,D)}. \tag{12}$$

The circuit operates under boundary conditions when the static gain in CCM and DCM is equal. In this mode, the converter exhibits characteristics of both configurations.

Using this assumption, positive and negative values of the square root are given as

$$Q_{1\text{BCM}} = \frac{n_T\left[n_T + \sqrt{n_T^2 - 16\overline{i_{R2}}\cdot n_T\cdot(n_{La}+1)}\right](n_{La}+1)}{2\cdot n_T + \left[n_T + \sqrt{n_T^2 - 16\overline{i_{R2}}\cdot n_T\cdot(n_{La}+1)}\right]\cdot n_{La}}. \tag{13}$$

Fig. 5(c) illustrates the output characteristics of the converter across multiple duty cycles, and parameterized load currents. The output characteristics of a converter define the behavior of the output voltage and current under varying operating conditions. In addition, they serve as a valuable tool for evaluating the converter's performance, efficiency, and stability across different scenarios.

## V. Stress Analysis of Forward in CCM

Based on the previous analysis of the proposed converter operating in CCM and forward mode, equations where developed to evaluate the stresses on various components.

The average current in $L_1$ corresponds to the load current, while the rms current is given by

$$I_{L1_{\text{rms}}} = \frac{I_{R2}}{2\,D\cdot n_{La}+1}\cdot\sqrt{2\,D\cdot(n_{La}+1)^2+1-2\,D}. \tag{14}$$

The rms current in $L_2$ is expressed as

$$I_{L2_{\text{rms}}} = I_{R2}\cdot\frac{\sqrt{1-2\,D}}{2\,D\cdot n_{La}+1}. \tag{15}$$

The maximum voltage in primary switches is same as source, and the rms current is given as

$$I_{Sp_{\text{rms}}} = I_{R2}\cdot\frac{n_T\,(n_{La}+1)\cdot\sqrt{D}}{2\,D\cdot n_{La}+1} + \frac{V_1\cdot\sqrt{D^3}}{2\,L_p\cdot f_s} \tag{16}$$

where $L_p$ denotes the magnetizing inductance of the transformer.

The maximum reverse voltage of primary diodes is same as source, and rms current as follows:

$$I_{Dp_{\text{rms}}} = \frac{V_2}{4\,L_p\cdot f_s}\cdot\frac{(2\,D\cdot n_{L_a}+1)\cdot\sqrt{D}}{n_T\,(n_{La}+1)}. \tag{17}$$

On the secondary side, the maximum voltage is determined by the transformation ratio of the transformer and the coupled

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

inductor and is expressed as

$$V_{D1_{\max}} = V_1 \cdot n_T + (V_1 \cdot n_T - V_2) \cdot n_{La}. \tag{18}$$

The rms current in secondary diodes is

$$I_{D1_{\text{rms}}} = \frac{I_{L2} \cdot \sqrt{1-2\,D}}{8\,D \cdot n_{La} + 1}. \tag{19}$$

For output capacitors, the most relevant parameter is output voltage, the rms current is given as follows:

$$I_{C2_{\text{rms}}} = I_{R2} \cdot \frac{n_{La}}{2D \cdot n_{La} + 1} \cdot \sqrt{2D \cdot (1-2D)}. \tag{20}$$

The average and rms currents in secondary of the transformers are the same as those the diodes. In primary, the rms current is

$$I_{Tp_{\text{rms}}} = \sqrt{\left[\frac{I_{R2} \cdot n_T(n_{La}+1)}{2D \cdot n_{La}+1} + \frac{V_1 \cdot D}{2L_p \cdot f_s}\right]^2 \left[\frac{V_1 \cdot D}{2L_p \cdot f_s}\right]^2} \cdot \sqrt{D}. \tag{21}$$

## VI. Operations Analysis of Flyback in CCM

In flyback mode, energy is transferred from $V_2$ to $V_1$. To represent $V_1$ as a load, a resistor is used in place of the voltage source, while an ideal voltage source is assigned to $V_2$.

To facilitate energy transfer from the secondary to the primary, the magnetizing inductances of the transformers function as coupled inductors, resulting in an operation similar to an interleaved flyback converter. This mode is achieved through the implementation of an auxiliary circuit composed of $L_1$, $C_b$, $D_{b1}$, and $D_{b2}$, which serves to create a low-pass filter for the secondary side. This configuration not only provides a stabilized voltage to the secondary but also establishes a current source characteristic when viewed from the perspective of source $V_2$. As a result, the circuit exhibits distinctive properties well-suited for flyback mode operation while also drawing a low-ripple current from the source. This feature is particularly advantageous for applications incorporating advanced battery technologies.

The conduction mode is analyzed by examining the current through the transformer windings, which now function as two coupled inductors. In CCM, during all steps, the load is supplied by one or both inductors simultaneously. $L_1$ serves as an auxiliary circuit, storing energy and discharging into capacitor $C_b$. No current flows through inductor $L_2$.

Fig. 10 illustrates all the steps involved in flyback CCM operation, while Fig. 7 depicts the key waveforms associated with this mode. The key idealized steady-state waveforms of the converter operating in flyback mode under DCM are shown in Fig. 8.

The static gain can be calculated using the average voltage across the coupled inductors. The average current in the magnetizing inductance of the transformer must be calculated using primary and secondary current. This average voltage is presented as

$$V_{Ls1}(t) = \frac{1}{T_s}\left(\int_0^{\Delta t_1} 2\,D \cdot {V_2}^2 dt + \int_0^{\Delta t_2} -\frac{V_1}{n'_T} dt\right). \tag{22}$$

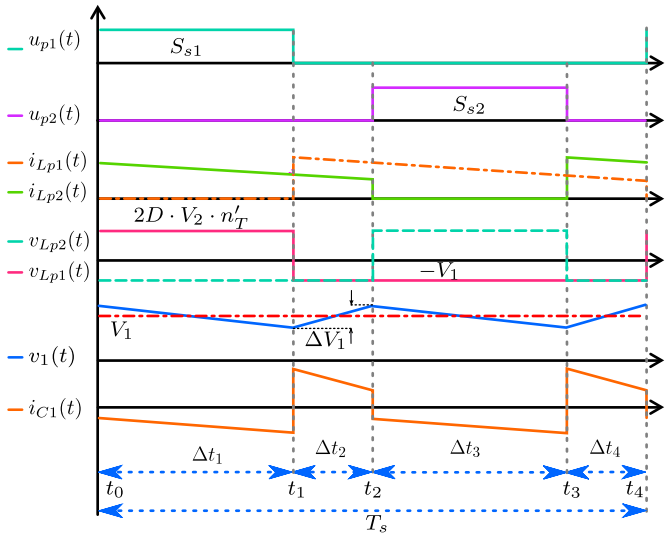



Fig. 7. Idealized waveforms of flyback in CCM.

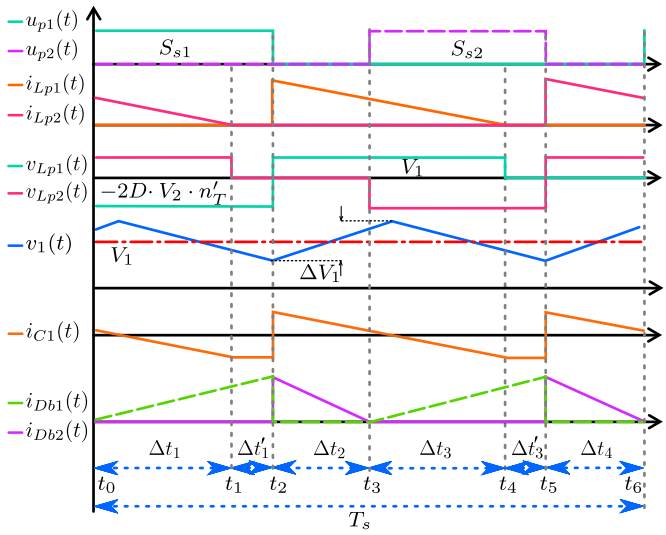



Fig. 8. Idealized waveforms of flyback in DCM.

Assuming

$$n'_T = \frac{1}{n_T}. \tag{23}$$

Energy is stored during $\Delta t_1$ and transferred to the load during $\Delta t_2$. The total period is denoted as $T_s$, corresponding to the current ripple frequency, as illustrated in Fig. 7.

Given that the average voltage across the inductor is zero, the static gain can be derived as shown in the following equation:

$$Q_{2\text{CCM}} = \frac{V_1}{V_2} = \frac{2\,D^2 \cdot n'_T}{1-D}. \tag{24}$$

The voltage gain curves corresponding to CCM are presented in Fig. 9(a). The flyback mode curves are compared with the quadratic flyback (F.q.) topology, considering a transformation ratio of $n'_T = 4$ and multiple transformer transformation ratios. Fig. 9(b) illustrates the static gain in flyback mode under DCM conditions as a function of $D$, for various parameterized load

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

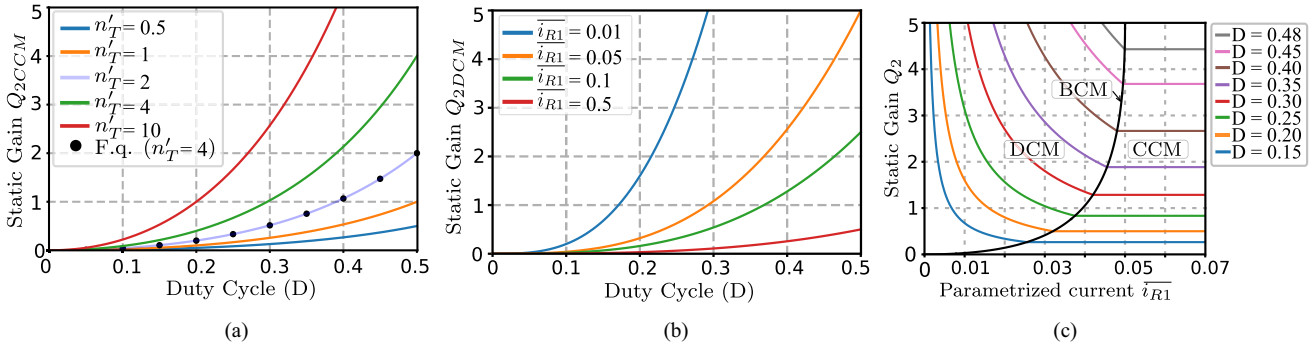



Fig. 9. Static gain of flyback mode. (a) CCM versus F.q. with $n'_T = 4$. (b) DCM. (c) Output characteristics.

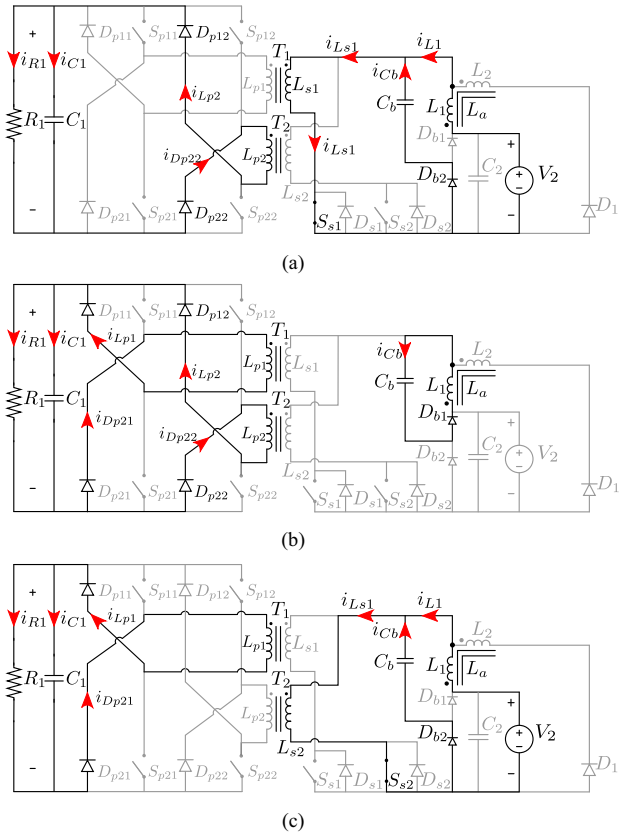



Fig. 10. Steps of flyback in CCM. (a) stage I $[t_0 - t_1]$, (b) stage II $[t_1 - t_2]$, and (c) stage III $[t_2 - t_3]$.

current values. Fig. 9(c) depicts the output characteristics, showing the static gain as a function of parameterized load current for CCM, DCM, and BCM across multiple values of $D$, highlighting the operating regions and their boundaries.

Compared to F.q., the proposed topology achieves double static gain. When analyzing the converter's behavior as a function of $n'_T$, it becomes evident that the converter can function as either a step-up or step-down converter, depending on the transformation ratio.

## VII. Operations Analysis of Flyback in DCM

In DCM, the converter operates through six distinct steps, during which the two inductors never supply the load simultaneously, as illustrated in the idealized waveforms in Fig. 8. Fig. 11 illustrates the additional steps involved in flyback mode under DCM operation. The first and third steps remain the same as in CCM, while two additional steps, A and B, are introduced afterward.

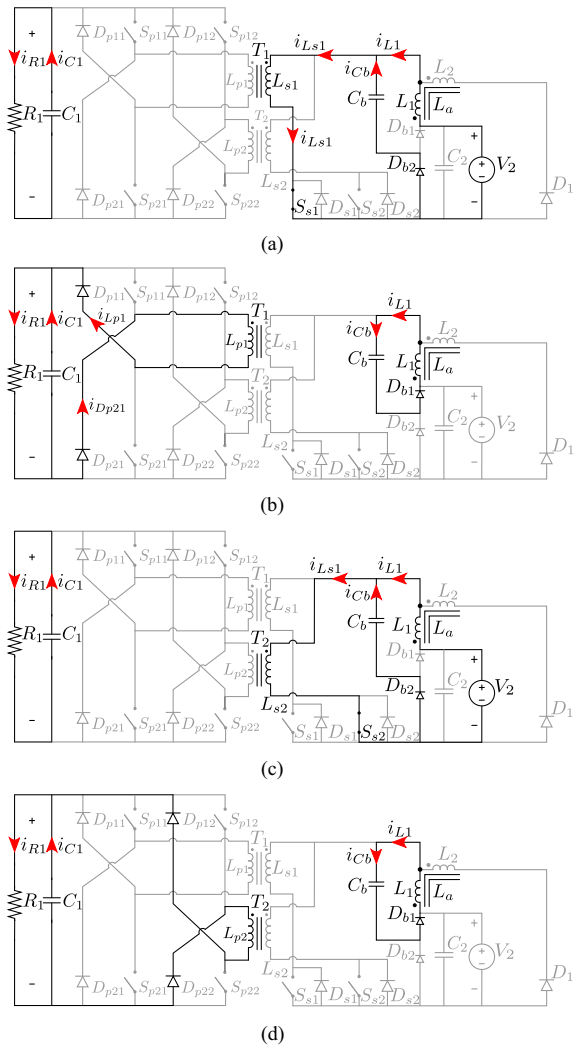



Fig. 11. Additional steps of flyback in DCM. (a) stage II $[t_1 - t_2]$, (b) stage III $[t_2 - t_3]$, (c) stage V $[t_4 - t_5]$, and (d) stage VI $[t_5 - t_6]$.

The average voltage applied on the primary is $2\,D \cdot V_2$. The peak current in diodes is equal to the inductor current. Using

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

this assumption, peak value of inductor current is given as

$$I_{D_{p12\text{pk}}} = I_{Lp_{\text{pk}}} = \frac{2\,D^2 \cdot V_2 \cdot T_s}{L_{s1} \cdot n'_T}. \tag{25}$$

Using (25), the time required for complete demagnetization of the inductors is given as

$$t_d = \frac{2\,D^2 \cdot V_2 \cdot T_s \cdot n'_T}{V_1}. \tag{26}$$

By applying the conservation law, the static gain is determined based on the energy balance. From this assumption, the energy delivered by the source is a product of input voltage, current, and $T_s$, while the energy consumed by the load is defined as

$$W_{V1} = V_1 \cdot \left( \frac{I_{D_{p12\text{pk}}}}{2} \cdot \frac{t_d}{T_s} + \frac{I_{D_{p11\text{pk}}}}{2} \cdot \frac{t_d}{T_s} \right) \cdot T_s. \tag{27}$$

The parameterized current in the load $I_{R1}$ is given as

$$\overline{i_{R1}} = \frac{I_{R1} \cdot L_{s1} \cdot f_s}{2\,D \cdot V_2}. \tag{28}$$

The static gain by parameterized current in the load is

$$Q_{2\text{DCM}} = \frac{2\,D^3}{\overline{i_{R1}}}. \tag{29}$$

Fig. 9(b) presents the static gain by variations of $D$, along with multiple curves of the parameterized current.

In BCM, the peak current equals the ripple, and the current in the secondary of the coupled inductors is given as

$$I_{Ls1_{\text{pk}}} = \frac{I_{R1} \cdot n'_T}{(1-D)}. \tag{30}$$

The boundary inductance is calculated as

$$L_{s1_{\text{BCM}}} = \frac{V_2}{I_{R1} \cdot f_s \cdot n'_T} \cdot 2\,D^2 \cdot (1-D). \tag{31}$$

The boundary duty ratio is given by

$$D_{\text{BCM}} = \frac{1}{2} \pm \frac{\sqrt{1 - 4\,\overline{i_{R1}} n'_T}}{2}. \tag{32}$$

Taking into account both the positive and negative solutions of the square root, the static gain for BCM can be expressed as follows:

$$Q_{2\text{BCM}} = \frac{\left(1 + \sqrt{1 - 4\,\overline{i_{R1}} \cdot n'_T}\right)^3}{4\,\overline{i_{R1}}}. \tag{33}$$

The output characteristics of the converter in flyback mode are presented in Fig. 9(c).

## VIII. Stress Analysis of Flyback in CCM

Based on the previous analysis of the proposed converter, the most critical stress parameters have been derived. In $L_1$, the rms current equals the mean current and is described as input current divided by two times $D$. In $L_{s1}$, the mean current is half of the current in $L_1$, and the rms is given as

$$I_{Ls1_{\text{rms}}} = I_{V2} \cdot \frac{\sqrt{D}}{4\,D^2}. \tag{34}$$

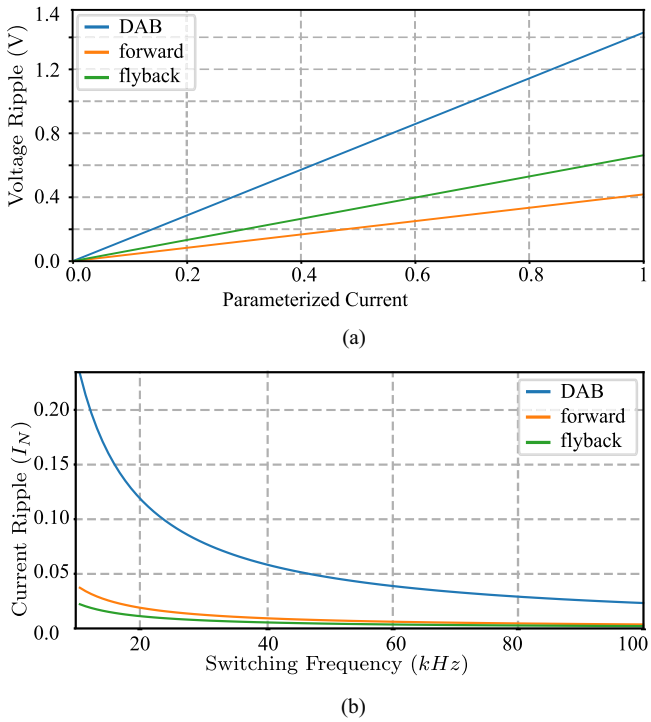



Fig. 12. Comparison of ripple between DAB and forward/flyback topologies. (a) Output voltage ripple versus parameterized current. (b) Current ripple in $L_1$ versus switching frequency.

The average current flowing through the primary winding is equal to half the load current, while the rms value is expressed as follows:

$$I_{Lp1_{\text{rms}}} = \frac{I_{R1}}{2} \cdot \frac{1}{\sqrt{1-D}}. \tag{35}$$

The maximum voltage in secondary switch $S_s$ is

$$V_{Ss_{\text{max}}} = \frac{V_1}{n'_T} + V_2 \cdot (1 + 2\,D). \tag{36}$$

The average current in the switch is equivalent to the inductor current. Consequently, the rms current in $S_s$ is expressed as

$$I_{Ss_{\text{rms}}} = I_{Ls1_{\text{rms}}} = I_{V2} \cdot \frac{D}{4\,D^2}. \tag{37}$$

The maximum reverse voltage in diodes $D_p$ is

$$V_{Dp_{\text{max}}} = \frac{V_2}{2} + V_2 \cdot n'_T \cdot D. \tag{38}$$

The rms current in the primary diodes is identical to that in the primary windings and is expressed as

$$I_{Dp_{\text{rms}}} = I_{Lp1_{\text{rms}}} = \frac{I_{R1}}{2} \cdot \frac{1}{\sqrt{1-D}}. \tag{39}$$

In the auxiliary circuit, the maximum reverse voltage in diodes is

$$V_{Db1_{\text{max}}} = V_{Db2_{\text{max}}} = V_2. \tag{40}$$

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

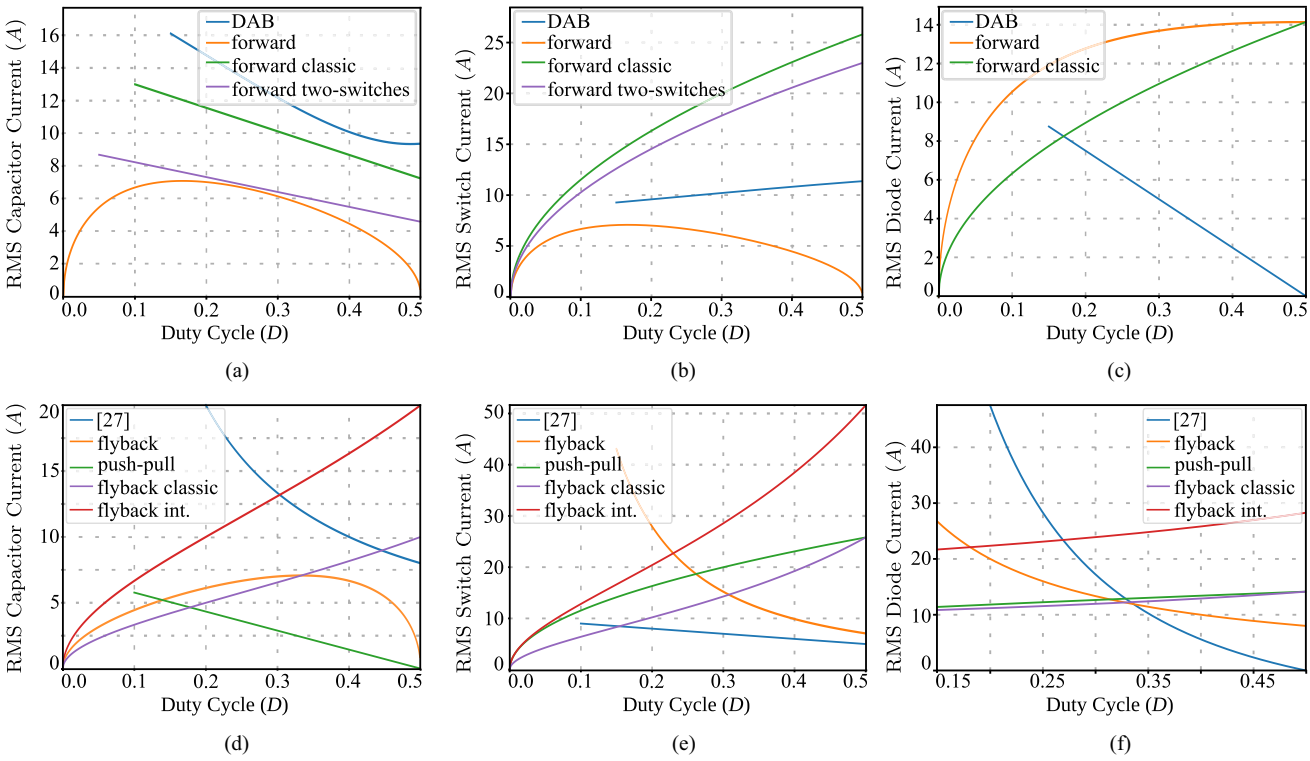



Fig. 13. RMS current comparison between forward/flyback and other converters. (a) Forward capacitor current. (b) Forward switch current. (c) Forward diode current. (d) Flyback capacitor current. (e) Flyback switch current. (f) Flyback diode current.

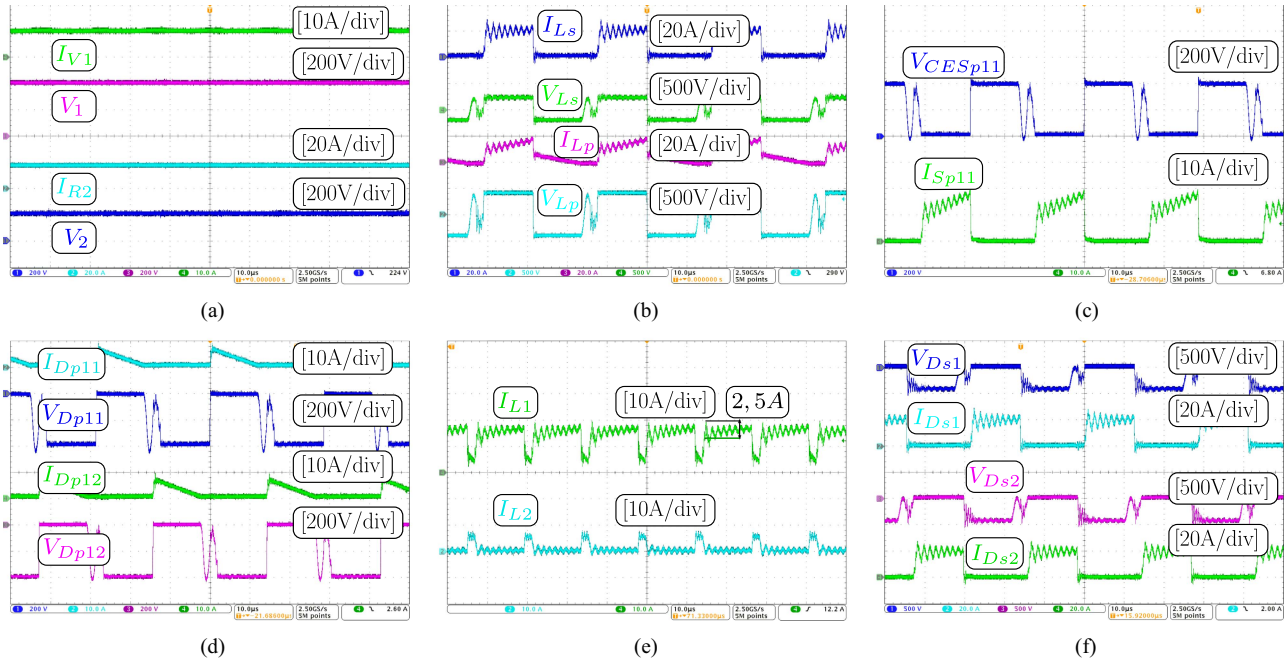



Fig. 14. Experimental waveforms for forward CCM operation. (a) Input and output current and voltage at nominal power of 3.5 kW. (b) Voltage and current measured in the primary and secondary of the transformers. (c) Voltage across the collector and emitter of the IGBT, along with the corresponding current. (d) Voltage and current in the primary diodes. (e) Current in both windings of the coupled inductor $L_a$. (f) Voltage and current measured in the secondary diodes.

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON WITH THE EXISTING STATE-OF-THE-ART AMONG SIMILAR BIDIRECTIONAL CONVERTERS

| Reference | Topology | $S$ | $D$ | $T_r$ | $L$ | $C$ | Modulation Technique | Input/Output Voltage ($V$) | Driving Circuit | Current ripple ($V_2$) | $f_s$ (kHz) | $P$ (kW) | $\eta$ (%) | Switching Devices |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| [17] | forward-flyback | 6 | 0 | 1 | 1 | 5 | PWM | 400/24−55 | Intermediate | Moderate | 40 | 0.5 | ∼ 94 | Si |
| [18] | DAB | 8 | 0 | 1 | 2 | 3 | PWM/PPS | 18−28/300 | complex | Small | 50 | 1 | ∼ 96 | Si |
| [19] | forward-flyback | 4 | 0 | 1 | 0 | 3 | PWM | 15/150 | Simple | Large | 50 | 0.3 | ∼ 92 | Si |
| [20] | CFBRC | 8 | 0 | 1 | 2 | 3 | PSM | 160−240/400 | complex | Small | 100 | 1.6 | ∼ 96 | Si |
| [21] | H5-CLLCBC | 9 | 0 | 2 | 3 | 5 | PFM | 230/420 | complex | Large | 82 | 1 | 98 | SiC |
| [22] | 4-phase flyback | 8 | 0 | 1 | 0 | 1 | PWM | 60/330 | Simple | Large | 20 | 6 | - | Si |
| [23] | 2-Switch flyback | 2 | 4 | 1 | 2 | 4 | PWM | 12/180 | Simple | Large | 50 | 0.05 | 90 | Si |
| [24] | IAC-BFC | 4 | 0 | 2 | 2 | 3 | PWM | 48/220 | Intermediate | Small | 100 | 0.3 | - | - |
| [25] | 2-phase BHB-CLL | 8 | 0 | 2 | 2 | 7 | PFM | 300/260 | complex | Large | 160 | 1.2 | ∼ 98 | - |
| [26] | BRCF-HB | 4 | 0 | 1 | 3 | 3 | PWM | 24/200 | Intermediate | Small | 50 | 0.2 | ∼ 95 | Si |
| [27] | DAB/Push-Pull | 8 | 0 | 1 | 1 | 2 | PWM | 42/4.2 | complex | Small | 50 | 0.021 | ∼ 85 | Si |
| [28] | 2-switch forward | 4 | 4 | 1 | 0 | 2 | PWM | 12/48 | Simple | Large | 100 | 0.02 | ∼ 88 | Si |
| Proposed | forward/flyback | 6 | 7 | 2 | 1 | 3 | PWM | 400/120−200 | Simple | Small | 35 | 3.5/1 | > 92/90 | Si/SiC |

($S$) switch, ($D$) diode, ($T_r$) transformer, ($L$) inductor, ($C$) capacitor, PWM, plus phase shift (PPS), phase-shift modulation (PSM), pulse–frequency modulation (PFM).

The average current in the diodes $D_b$ of the auxiliary circuit is

$$I_{Db1_{\text{avg}}} = I_{Db2_{\text{avg}}} = \frac{I_{V2}}{2} \cdot \frac{1-2\,D}{D}. \tag{41}$$

The rms current in $D_{b1}$ is

$$I_{Db1_{\text{rms}}} = \frac{I_{V2}}{2} \cdot \frac{\sqrt{1-2\,D}}{D}. \tag{42}$$

The rms current for $D_{b2}$ is

$$I_{Db2_{\text{rms}}} = \frac{I_{V2}}{4} \cdot \frac{(1-2\,D)\sqrt{2\,D}}{D^2}. \tag{43}$$

The current through capacitors $C_1$ and $C_2$ is presented below, along with the maximum voltage across $C_{b1}$

$$I_{C1_{\text{rms}}} = \frac{I_{R1}}{(1-D)} \cdot \sqrt{\frac{(1-2\,D)\cdot D}{2}} \tag{44}$$

$$I_{Cb1_{\text{rms}}} = \frac{I_{V2}}{2\,D} \cdot \sqrt{\frac{1-2\,D}{2\,D}} \tag{45}$$

$$V_{Cb1_{\text{max}}} = V_2 \cdot 2\,D + \frac{\Delta V_{Cb1}}{2}. \tag{46}$$

## IX. COMPARISON WITH ISOLATED BIDIRECTIONAL CONVERTERS

A comparative analysis was conducted between the forward/flyback converter and other converters with similar characteristics to assess their respective advantages and limitations. In addition, a comparison of various newly developed bidirectional converters is presented, considering factors such as component count, modulation, PWM signal complexity, current ripple, and other relevant aspects, as summarized in Table I. The voltage ripple is influenced by variations in the load current. Fig. 12(a) demonstrates the output ripple as a function of the parameterized load current. Furthermore, the current ripple is another critical factor, which is illustrated in Fig. 12(b) as a function of the switching frequency.

To compare the current stress in capacitors, switches, and diodes, tree graphs were generated for each mode, as described in Fig. 13. These graphs offer a comparison of the proposed topology with some of the most widely adopted converters in the industry, including DAB, isolated voltage step-up and step-down converters, and current-fed topologies. It is worth noting that the secondary side of the proposed converter exhibits low ripple current, making it well-suited for bidirectional energy storage applications. Key advantages of the proposed converter include its simple and modular design, a reduced number of passive components, low rms stresses, and the ability to be effectively utilized in medium- to high-power applications.

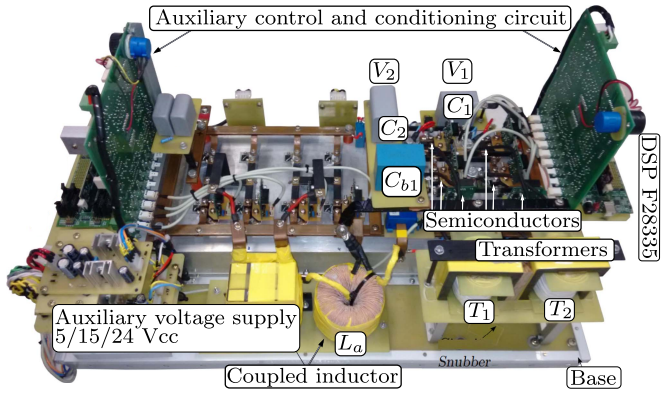



Fig. 15. Prototype evaluated, forward/flyback converter.

## X. EXPERIMENTAL VERIFICATION

The design and dimensioning of the converter were carefully developed to align with the criteria and specific requirements of applications that demand energy storage via batteries, galvanic isolation, and bidirectionality. These applications include RESS, dc microgrids, and EVs. In this context, the forward mode facilitates the transfer of dc energy from a dc bus (∼400 V) or, for instance, from a front-end power factor correction stage to a battery.

In flyback mode, the converter was intentionally designed to operate in DCM for several key reasons. First, it maintains the same voltage gain as in forward mode, where the source $V_1$ could represent a dc bus in microgrid applications. In addition, DCM operation provides a very fast dynamic response and ensures stability under all operating conditions. In contrast, CCM may lead to diode reverse recovery issues, potentially causing noise, electromagnetic interference, and additional losses. Furthermore, operating in DCM allows the transistors to switch on with zero current, minimizing switching losses. Finally, this approach enables the use of a smaller transformer, contributing to a more compact and efficient design.

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

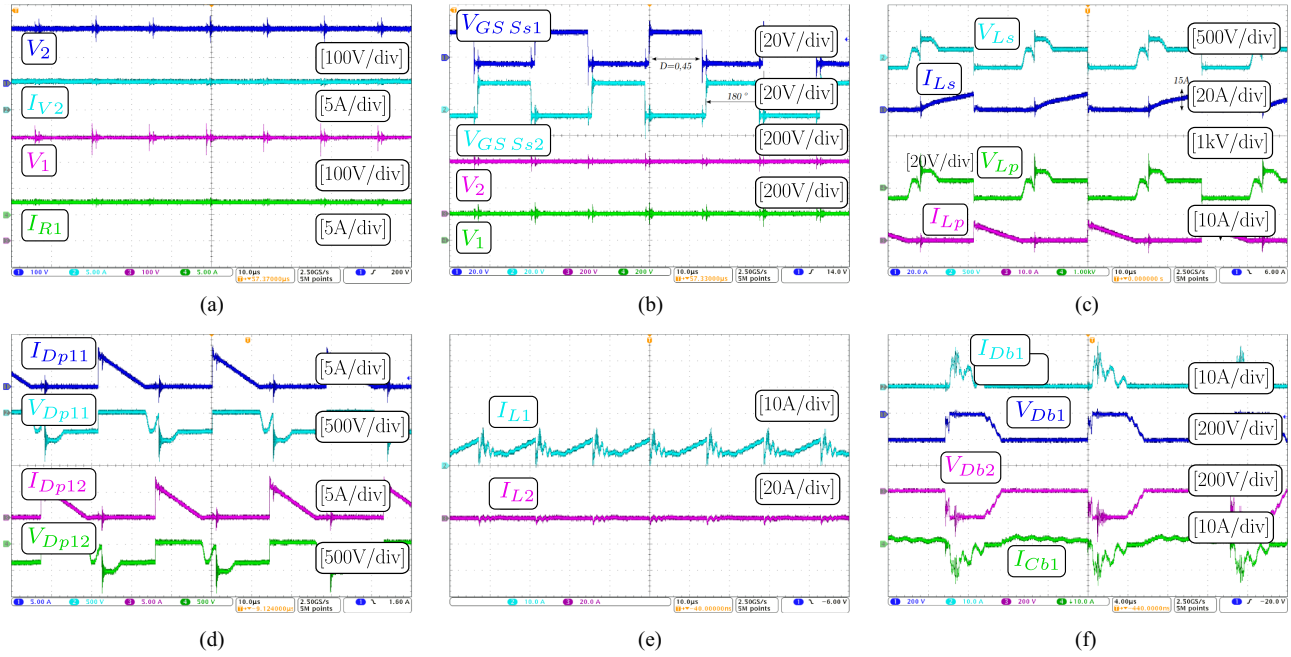



Fig. 16. Experimental waveforms for flyback DCM operation. (a) Input and output current and voltage at a nominal power of 1 kW. (b) Voltage measured at the gate and source of the MOSFETs, along with input and output voltages. (c) Voltage measured in the primary and secondary windings of coupled inductors $T_1$ and $T_2$. (d) Voltage and current in the primary diodes. (e) Current through both windings of the coupled inductor $L_a$. (f) Voltage and current in the auxiliary circuit, along with the current of $C_b$.

TABLE II
COMPONENTS USED IN THE PROTOTYPE

| Part | Description |
|---|---|
| flyback MOSFET $S_s$ (SiC) | 2 APT40SM120J (Microsemi, SOT-227) |
| flyback diodes $D_s$ (SiC Schottky) | 2 MSC020SDA120B (Microsemi, TO-247) |
| forward IGBT $S_p$ (Si) | 4 APT100GT60JR (Microsemi, SOT-227) |
| forward diodes $D_p$ (SiC Schottky) | 4 STPSC20H12D (ST, TO-220) |
| Auxiliary diodes $D_b$, $D_1$ (SiC Schottky) | 3 IDW30G65C5 (Infineon, TO-247) |
| Output capacitor $C_1$, $C_2$ (Film) | 2 MKP1848650084Y5 (VISHAY, 50 $\mu$F / 800 V) |
| Auxiliary capacitor $C_b$ (Film) | 1 B32678G4356K (Epcos, 35 $\mu$F / 450 V) |
| Coupled inductor ($L_a$): $L_1$, $L_2$ | Sendust $n_1$=18 / $n_2$=34 turns (Magmattec, 4 //: MMTS60T7713) |
| Transformer $T_1$, $T_2$ | Ferrite $n_p$=20 / $n_s$=11 turns (Thornton, 2 // NEE-80/38/20) |
| DSP (texas instruments) | TMS320F28335 (Delfino experimenter kit) |

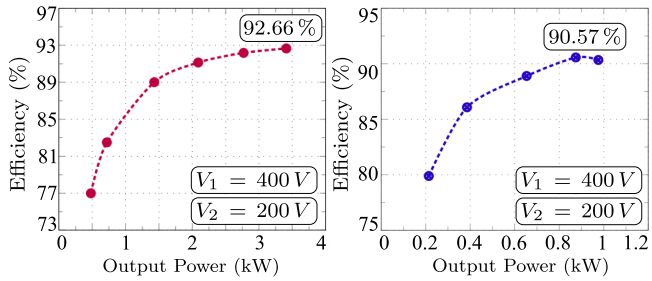



Fig. 17. Efficiency curve for 400 V/200 V ($V_1$/$V_2$). (a) Forward mode. (b) Flyback mode.

To validate the proof of concept, an experimental prototype of the interleaved forward/flyback converter was meticulously designed and assembled in the laboratory, adhering to the key application guidelines. The converter operates with a maximum output power of 3.5 kW, in forward mode under CCM, with a primary-side magnetizing inductance of $Lm_p = 595\,\mu$H, a turns ratio of $n_T = 0.55$, a ripple current of 14%, resulting in $L_1 = 86\,\mu$H (4), and $n_{La} = 1.88$. The switching frequency is set to 35 kHz, with a maximum duty cycle of 0.45 for both operating modes. The input voltage ($V_1$) is fixed at 400 V, while the output voltage varies between 120 and 200 V, maintaining a voltage ripple ($\Delta V_2$) of 0.5%. In flyback mode, the converter operates in DCM, featuring a secondary-side magnetizing inductance of $Lm_s = 180\,\mu$H, an output power of 1 kW at 400 V, and a maximum voltage ripple ($\Delta V_1$) of 0.1%. The corresponding parameterized current is given by $\overline{i_{R1}} = 0.09$ (28), with an input voltage ($V_2$) of 200 $V$. Fig. 15 presents a photograph of the prototype of the proposed converter. The hardware parameters are detailed in Table II.

Two experimental tests were conducted to validate the behavior of the forward and flyback modes. These tests were performed at nominal power to highlight the key waveforms and confirm the previously proposed theory.

Fig. 14 presents the experimental results from tests conducted in forward mode under CCM, highlighting the key waveforms that validate the converter's operation. Similarly, Fig. 16 demonstrates the experimental results from tests performed in flyback mode under DCM.

To assess the efficiency of both converters, tests were performed using a Tektronix, $PA4000$ power analyzer. In forward mode, the maximum efficiency achieved was 92.66%, corresponding to an output power of 3.412 kW, as illustrated in the graph presented in Fig. 17(a) .

The tests were also conducted in flyback mode, with the maximum power point achieved at 874.4 W and an efficiency of 90.57%. At the designated power point, the maximum efficiency reached 90.34%. Beyond this point, efficiency gradually

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.




(a) (b)

Fig. 18. Details of soft-switching in forward mode in CCM. (a) ZVS turn-ON. (b) turn-OFF waveform.

declined due to increased circuit losses. The power transferred in this mode was constrained by the voltage gain delivered to the load, as the converter operates in DCM. Efficiency degradation was also observed due to the necessity of additional dissipative resistor–capacitor–diode clamping, which is used to limit the overvoltage on the MOSFETs caused by the leakage inductance of the transformer windings. The efficiency curve in flyback mode is tested, as shown in 17(b).

A detailed analysis of the converter operating at nominal power with ZVS is provided in Fig. 18(a) and (b). The experiment was conducted with the forward converter operating in CCM. To verify the ZVS condition, the voltage and current across one of the primary switches were measured. Fig. 18(a) illustrates the moment when the switch begins to conduct, while Fig. 18(b) depicts the moment when the switch ceases conduction.

## XI. Conclusion

In this study, we investigated a novel interleaved forward/flyback bidirectional dc–dc converter, designed to address the growing demands of modern energy storage systems, EVs, and dc microgrids. The proposed topology integrates coupled inductors and interleaved operation, achieving a wide voltage gain range, reduced current ripple, and galvanic isolation, while maintaining high efficiency and bidirectional power flow capability. Notably, the implementation of ZVS in forward mode markedly reduces switching losses and electromagnetic interference, further augmenting the converter's performance.

The operational principles of the converter were thoroughly examined across continuous (CCM), discontinuous (DCM), and boundary (BCM) conduction modes, with detailed analytical models developed to characterize voltage and current stresses on key components. Experimental results of the developed prototype confirmed its bidirectional capability and high efficiency in both forward and flyback modes, outperforming equivalent topologies with similar characteristics. These findings validate the theoretical analysis and underscore the practical applicability of the converter. The converter's ability to operate with low ripple current and its modular design make it particularly suitable for integration into advanced battery systems and renewable energy applications where reduced complexity and cost are paramount.

Despite these advancements, certain challenges persist, particularly in optimizing the tradeoffs between efficiency, power density, and cost, especially under wide input/output voltage variations. Careful consideration should be given to the design of transformers, coupled inductors, and the overall layout to effectively minimize parasitic effects and mitigate the high resonance frequencies observed in the experiment. In comparison to other bidirectional converters, such as the DAB, the proposed converter in forward mode exhibits lower rms current in capacitors and switches, albeit higher in diodes. Conversely, in flyback mode, the converter presents reduced rms current in capacitors, increased rms current in switches, and the rms current in diodes depends on duty cycle operation point.

Future endeavors will be dedicated to refining control strategies, optimizing magnetic elements, further reducing component stresses, and exploring the integration of wide-bandgap semiconductors to enhance efficiency and power density. The proposed converter represents a significant advancement in power conversion technology, achieving bidirectionality through transformers without the need for additional windings or external circuits. This innovation offers a robust and scalable solution for next-generation power systems.

## Acknowledgment

The authors would like to thank UDESC and FURB Universities.

## References

[1] D. Das, V. M. Hrishikesan, C. Kumar, and M. Liserre, "Smart transformer-enabled meshed hybrid distribution grid," *IEEE Trans. Ind. Electron.*, vol. 68, no. 1, pp. 282–292, Jan. 2021.

[2] J. R. Rao and S. Venkateshwarlu, "Soft-switching dual active bridge converter-based bidirectional on-board charger for electric vehicles under vehicle-to-grid and grid-to-vehicle control optimization," *J. Eng. Appl. Sci.*, vol. 71, no. 1, 2024, Art. no. 49.

[3] F. Mwasilu, J. J. Justo, E.-K. Kim, T. D. Do, and J.-W. Jung, "Electric vehicles and smart grid interaction: A review on vehicle to grid and renewable energy sources integration," *Renewable Sustain. Energy Rev.*, vol. 34, pp. 501–516, 2014.

[4] V. Sultan, A. Aryal, H. Chang, and J. Kral, "Integration of EVS into the smart grid: A systematic literature review," *Energy Informat.*, vol. 5, no. 1, 2022, Art. no. 65.

[5] K. M. Muttaqi, M. R. Islam, and D. Sutanto, "Future power distribution grids: Integration of renewable energy, energy storage, electric vehicles, superconductor, and magnetic bus," *IEEE Trans. Appl. Supercond.*, vol. 29, no. 2, Mar. 2019, Art. no. 3800305.

[6] S. Chavhan, V. Mohale, and M. Kumbhar, "A review of recent advancements and challenges in battery energy storage system (BESS)," in *Proc. 1st Int. Conf. Innov. Sustain. Technol. Energy Mechatron. Smart Syst.*, 2024, pp. 1–6.

[7] T. Instruments, "Isolated bidirectional DC/DC in power conversion system (PCS)," Texas Instruments, Dallas, TX, Tech. Rep. 2, 2024.

[8] X. Pan, H. Li, Y. Liu, T. Zhao, C. Ju, and A. K. Rathore, "An overview and comprehensive comparative evaluation of current-fed-isolated-bidirectional DC/DC converter," *IEEE Trans. Power Electron.*, vol. 35, no. 3, pp. 2737–2763, Mar. 2020.

[9] B. Zhao, Q. Song, W. Liu, and Y. Sun, "Overview of dual-active-bridge isolated bidirectional DC–DC converter for high-frequency-link power-conversion system," *IEEE Trans. Power Electron.*, vol. 29, no. 8, pp. 4091–4106, Aug. 2014.

[10] S. Li, X. Yuan, Z. Wang, K. Wang, Y. Zhang, and X. Wu, "A unified optimal modulation strategy for DAB converters to tradeoff the backflow power reduction and all ZVS in the full operating range," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 11, no. 6, pp. 5701–5723, Dec. 2023.

[11] S. Hamada and M. Nakaoka, "A novel zero-voltage and zero-current switching PWM DC-DC converter with reduced conduction losses," *IEEE Trans. Power Electron.*, vol. 17, no. 3, pp. 413–419, May 2002.

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.

[12] H. Feng, D. Xu, and Y. Lee, "Comparison of interleaving methods of two-transistor forward converters," in *Proc. IEEE Int. Conf. Power Electron. Drive Syst.*, 1999, pp. 632–637.
[13] B. Tamyurek and B. Kirimer, "An interleaved high-power flyback inverter for photovoltaic applications," *IEEE Trans. Power Electron.*, vol. 30, no. 6, pp. 3228–3241, Jun. 2015.
[14] H.-L. Cheng, Y.-N. Chang, C.-H. Chang, S.-Y. Hsieh, and C.-A. Cheng, "A novel high-power-factor AC/DC LED driver with dual flyback converters," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 7, no. 1, pp. 555–564, Mar. 2019.
[15] W. Li and X. He, "A family of isolated interleaved boost and buck converters with winding-cross-coupled inductors," *IEEE Trans. Power Electron.*, vol. 23, no. 6, pp. 3164–3173, Nov. 2008.
[16] K.-C. Tseng, J.-Z. Chen, J.-T. Lin, C.-C. Huang, and T.-H. Yen, "High step-up interleaved forward-flyback boost converter with three-winding coupled inductors," *IEEE Trans. Power Electron.*, vol. 30, no. 9, pp. 4696–4703, Sep. 2015.
[17] Y.-E. Wu and Y.-T. Ke, "A novel bidirectional isolated DC-DC converter with high voltage gain and wide input voltage," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 7973–7985, Jul. 2021.
[18] D. Sha, X. Wang, and D. Chen, "High-efficiency current-fed dual active bridge DC–DC converter with ZVS achievement throughout full range of load using optimized switching patterns," *IEEE Trans. Power Electron.*, vol. 33, no. 2, pp. 1347–1357, Feb. 2018.
[19] F. Zhang and Y. Yan, "Novel forward–flyback hybrid bidirectional DC–DC converter," *IEEE Trans. Ind. Electron.*, vol. 56, no. 5, pp. 1578–1584, May 2009.
[20] Y. Li, Y. Xing, Y. Lu, H. Wu, and P. Xu, "Performance analysis of a current-fed bidirectional LLC resonant converter," in *Proc. 42nd Annu. Conf. IEEE Ind. Electron. Soc.*, 2016, pp. 2486–2491.
[21] M. Zhou, D. Shu, and H. Wang, "An H5-bridge-based laddered *CLLC* DCX with variable DC link for PEV charging applications," *IEEE Trans. Power Electron.*, vol. 37, no. 4, pp. 4249–4260, Apr. 2022.
[22] T. Bhattacharya, V. S. Giri, K. Mathew, and L. Umanand, "Multiphase bidirectional flyback converter topology for hybrid electric vehicles," *IEEE Trans. Ind. Electron.*, vol. 56, no. 1, pp. 78–84, Jan. 2009.
[23] N. M. Mukhtar and D. D.-C. Lu, "A bidirectional two-switch flyback converter with cross-coupled lcd snubbers for minimizing circulating current," *IEEE Trans. Ind. Electron.*, vol. 66, no. 8, pp. 5948–5957, Aug. 2019.
[24] S. Pan et al., "A high-gain interleaved active-clamped bidirectional flyback converter," in *Proc. IEEE 7th Int. Elect. Energy Conf.*, 2024, pp. 3669–3674.
[25] X. Zhu, K. Liu, B. Zhang, and K. Jin, "Multiphase BHB-CLL resonant converter based on secondary-side VDR with automatic current sharing characteristic," *IEEE Trans. Ind. Electron.*, vol. 71, no. 2, pp. 1443–1455, Feb. 2024.
[26] T.-J. Liang and J.-H. Lee, "Novel high-conversion-ratio high-efficiency isolated bidirectional DC–DC converter," *IEEE Trans. Ind. Electron.*, vol. 62, no. 7, pp. 4492–4503, Jul. 2015.
[27] S. Yu, M. Q. Nguyen, and W. Choi, "A novel soft-switching battery charge/discharge converter with the zero voltage discharge function," *IEEE Trans. Power Electron.*, vol. 31, no. 7, pp. 5067–5078, Jul. 2016.
[28] N. M. Mukhtar and D. D.-C. Lu, "An isolated bidirectional forward converter with integrated output inductor-transformer structure," in *Proc. IEEE 4th Southern Power Electron. Conf.*, 2018, pp. 1–7.



**Robson Mayer** received the B.S. degree in electrical engineering from the Católica de Santa Catarina, Jaraguá do Sul, Brazil, in 2010, the M.Sc. degree in electrical engineering from the Universidade Regional de Blumenau (FURB), Blumenau, Brazil, in 2014, and the Ph.D. degree in electrical engineering from the Universidade do Estado de Santa Catarina (UDESC), Joinville, Brazil, in 2019.

He has been actively engaged in research in the field of power electronics with both the Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil, and UDESC. His efforts focus on power electronics, with specialization in dc–dc converters, inverters, power factor correction (PFC) converters, and rectifiers. His research interests include the design and control of power electronic converters, the application of power electronics in renewable energy systems and electric vehicles, energy efficiency, resonant dc–dc converters, and high-power density solutions.



**Luan Souza de Oliveira** received the master's degree in electrical engineering from Universidade do Estado de Santa Catarina (UDESC), Joinville, in 2025. He specializes in power processing and control strategies for electronic converters from UDESC, where he is currently working toward the Ph.D. degree in electrical engineering.

With a strong background in power electronics, he has dedicated his career to advancing control techniques, modulation strategies, and renewable energy applications. His work aims to contribute to the development of efficient and sustainable power electronic systems, particularly in the context of modern energy challenges. His research focuses on matrix converters and power processing for electric vehicles (EVs) with UDESC. His research interests include power electronics, power processing, and the control and modulation of power converters, striving to bridge innovative research with real-world solutions.



**Menaouar Berrehil El Kattel** was born in Terga, Ain Temouchent, Algeria, in 1978. He received the bachelor's degree in electrical engineering from the Djillali Liabes University of Sidi Bel Abbès, Sidi Bel Abbès, Algeria, in 2003, the master's degree in electrical engineering from the Regional University of Blumenau (FURB), Blumenau, Brazil, in 2013, and the Ph.D. degree in electrical engineering from the State University of Santa Catarina (UDESC), Florianópolis, Brazil, in 2018.

He is currently a Professor of Electrical Engineering with the Federal University of Ceará (UFC), Fortaleza, Brazil, and a Researcher in Power Electronics with the Federal University of Minas Gerais (UFMG), Belo Horizonte, Brazil. His research interests include dc–dc converters, ac–dc converters, soft switching, and power electronics applications in electric vehicles.



**Luiz Carlos Gili** received the bachelor's and master's degrees in electrical engineering from the Regional University of Blumenau, Blumenau, Brazil, in 2014 and 2016, respectively, and the Ph.D. degree in electrical engineering from the Federal University of Santa Catarina (UFSC), Florianópolis, Brazil, in 2024.

He is currently a Postdoctoral Research with Universidade do Estado de Santa Catarina (UDESC), Joinville, Brazil, focused on advanced modulation and control techniques for static converters. Since 2024, he has been a Senior Researcher with NIDEC Global Appliance, Joinville, Brazil. His research interests include with digital electronics, hardware/software integration in power electronics, and innovative energy conversion solutions, with a strong commitment to enhancing the efficiency and reliability of modern power systems.



**Sérgio Vidal Garcia Oliveira** (Senior Member, IEEE) was born in Lages, SC, Brasil, in 1974. He received the B.S. degree in electrical engineering from Universidade Regional de Blumenau (FURB), Blumenau, Brazil, in 1999 and the M.Sc. and Ph.D. degrees in electrical engineering from Universidade Federal de Santa Catarina (UFSC), Florianópolis, Brazil, in 2001 and 2006, respectively.

He is currently a Technological Development and Innovative Extension Productivity Grant Holder DT-2 (CNPq), He has been a Professor of Power Electronics with FURB in 2004 and UDESC in 2012. His research interests are integrated motor drives, solid-state transformers, power converters to electric traction systems, cybersecurity on power electronics, and design to reliability in power electronics.

Dr. Oliveira is currently a member of SOBRAEP—Brazilian Power Electronics Society, SBA—Brazilian Automatic Society, ABENGE—Brazilian Education in Engineering Association, IES—Industrial Electronics Society, PELS—Power Electronics Society, and IAS—Industry Applications Society.

Authorized licensed use limited to: Karl Raymond Roque. Downloaded on August 10,2026 at 11:51:00 UTC from IEEE Xplore. Restrictions apply.